



GCE A LEVEL MARKING SCHEME

**A LEVEL
CHEMISTRY
1410U40 – 1**

About this marking scheme

The purpose of this marking scheme is to provide teachers, learners, and other interested parties, with an understanding of the assessment criteria used to assess this specific assessment.

This marking scheme reflects the criteria by which this assessment was marked in a live series and was finalised following detailed discussion at an examiners' conference. A team of qualified examiners were trained specifically in the application of this marking scheme. The aim of the conference was to ensure that the marking scheme was interpreted and applied in the same way by all examiners. It may not be possible, or appropriate, to capture every variation that a candidate may present in their responses within this marking scheme. However, during the training conference, examiners were guided in using their professional judgement to credit alternative valid responses as instructed by the document, and through reviewing exemplar responses.

Without the benefit of participation in the examiners' conference, teachers, learners and other users, may have different views on certain matters of detail or interpretation. Therefore, it is strongly recommended that this marking scheme is used alongside other guidance, such as published exemplar materials or Guidance for Teaching. This marking scheme is final and will not be changed, unless in the event that a clear error is identified, as it reflects the criteria used to assess candidate responses during the live series.

WJEC GCE A LEVEL CHEMISTRY
PHYSICAL AND ORGANIC CHEMISTRY
GENERAL INSTRUCTIONS

Recording of marks

Examiners must mark in red ink.

One tick must equate to one mark (apart from the questions where a level of response mark scheme is applied).

Question totals should be written in the box at the end of the question.

Question totals should be entered onto the grid on the front cover and these should be added to give the script total for each candidate.

Marking rules

All work should be seen to have been marked.

Marking schemes will indicate when explicit working is deemed to be a necessary part of a correct answer.

Crossed out responses not replaced should be marked.

Credit will be given for correct and relevant alternative responses which are not recorded in the mark scheme.

Extended response question

A level of response mark scheme is used. Before applying the mark scheme please read through the whole answer from start to finish. Firstly, decide which level descriptor matches best with the candidate's response: remember that you should be considering the overall quality of the response. Then decide which mark to award within the level. Award the higher mark in the level if there is a good match with both the content statements and the communication statement. Award the middle mark in the level if most of the content statements are given and the communication statement is partially met. Award the lower mark if only the content statements are matched.

Marking abbreviations

The following may be used in marking schemes or in the marking of scripts to indicate reasons for the marks awarded.

cao = correct answer only

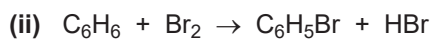
ecf = error carried forward

bod = benefit of doubt

1 (a) (i)

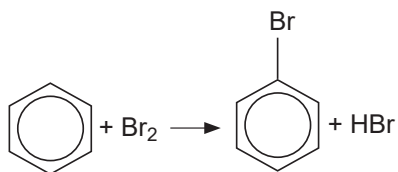
Reaction of benzene	Name of organic product	Name of catalyst
with bromine	bromobenzene	iron/iron(III) bromide/ aluminium bromide [1]
with ethanoyl chloride	phenylethanone [1]	iron(III) chloride/ aluminium chloride [1]

[3]



[1]

or

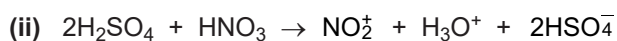


(iii) electrophilic substitution

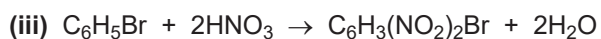
[1]

(b) (i) concentrated sulfuric acid [1]
concentrated nitric acid [1]

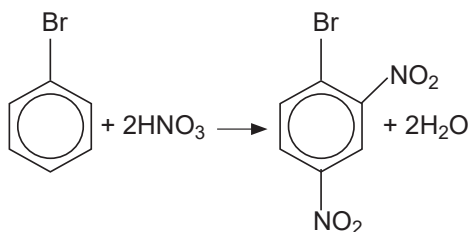
[2]



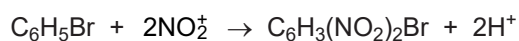
[2]



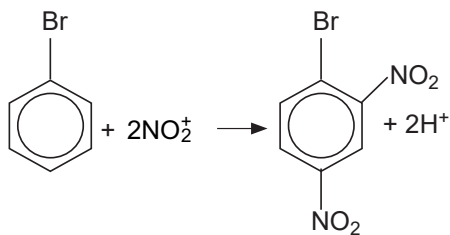
or



or



or



[1]

AVAILABLE
MARKS

2 (a) propanedioic acid [1]

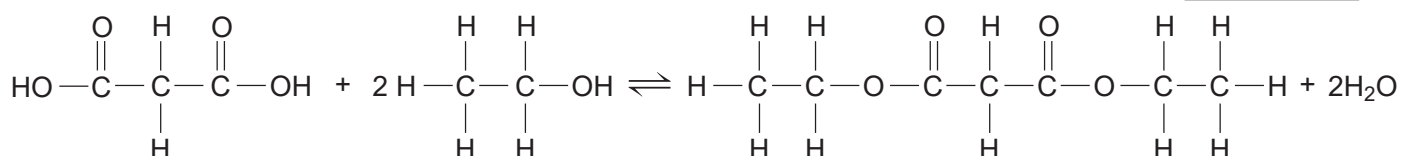


or

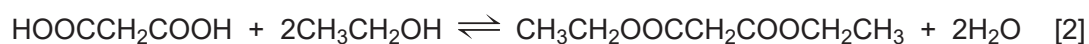


(ii) propanedioyl (di)chloride [1]

(d) (i)



or



(ii) van der Waals' forces and permanent dipole-dipole attractions
between molecules of ester [1]
stronger hydrogen bonds between molecules of acid [1] [2]



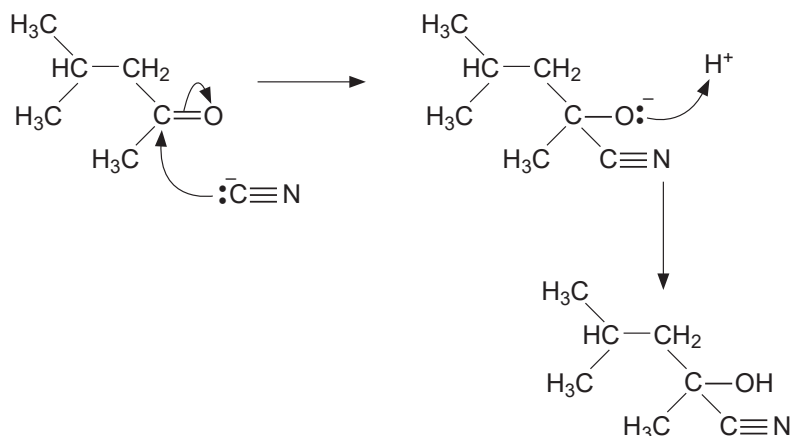
(ii) reduction [1]



AVAILABLE
MARKS

14

3 (a) (i)



3 curly arrows correct [1]

correct charges on ^-CN , O^- , H^+ [1]

lone pairs on C of ^-CN and O^- of intermediate [1]

[3]

(ii) nucleophilic addition

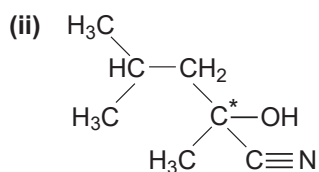
[1]

(iii) 4-methylpentan-2-one

[1]

(b) (i) an atom which has four different atoms or groups attached

[1]



[1]

(iii) CN^- attack equally probable from top or bottom/racemic mixture/equal concentrations of both optical isomers formed

[1]

(c) (i) 2-hydroxy-2,4-dimethylpentanoic acid

[2]

(ii) ammonia

[1]

(iii) ethanal

[1]

Section B

100

Total

110

AVAILABLE
MARKS

12

4 (a) (i) nitrobenzene

[1]

(ii) tin [1]
concentrated hydrochloric acid [1]

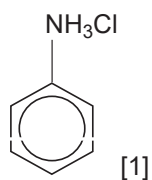
[2]

(iii) reduction

[1]

(iv)

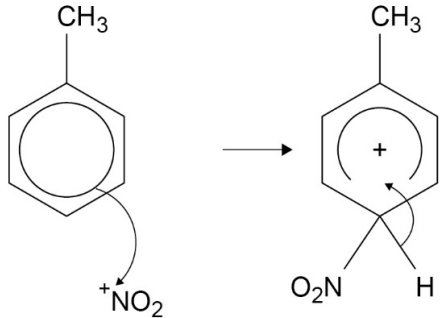
[2]

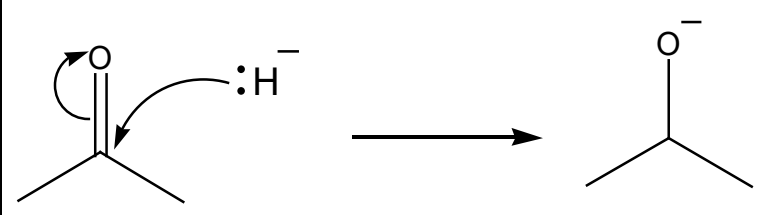


name: phenylammonium chloride [1]

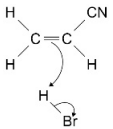
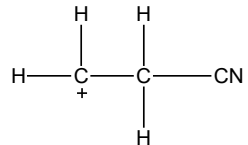
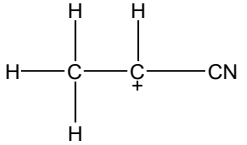
[2]

AVAILABLE
MARKS

Question	Answers	Additional Comments/Guidelines	Mark
05.1	 M4 Electrophilic substitution	of $^+\text{NO}_2$ M2 Arrow from ring (inside hexagon) to N or + on N M3 Arrow from C-H bond into hexagon	4 (4 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
05.2	 M4 Nucleophilic addition	M1 Arrow from lp on hydride to C M2 Arrow from C=O to O M3 Intermediate structure / Allow displayed or abbreviated structures Ignore any attempt to show further steps if correct Penalise further incorrect steps Apply list principle	4 (4 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
06.1	3-bromopropan <u>e</u> nitrile	Allow 3-bromopropan <u>e</u> -1-nitrile	1 (AO1)

Question	Answers		Additional Comments/Guidelines	Mark
06.2	This question is marked using Levels of Response. Refer to the Mark Scheme Instructions for Examiners for guidance.		Indicative Chemistry content	6 QER
	Level 3 5–6 marks	All stages are covered and each stage is generally correct and virtually complete. Answer is communicated coherently and shows a logical progression from Stage 1 to Stages 2 and 3.	Stage 1 Types of Isomers formed 1a CH_3CHBrCN 1b Exists as two Optical isomers / enantiomers	
	Level 2 3–4 marks	All stages are covered but stage(s) may be incomplete or may contain inaccuracies OR two stages are covered and are generally correct and virtually complete. Answer is communicated mainly coherently and shows a logical progression from Stage 1 to Stages 2 and 3.	Stage 2 Mechanism 2a 2 curly arrows 	
	Level 1 1–2 marks	Two stages are covered but stage(s) may be incomplete or may contain inaccuracies OR only one stage is covered but is generally correct and virtually complete. Answer includes isolated statements but these are not presented in a logical order.	2b Intermediate structure primary carbocation OR 	
	0 mark	Insufficient correct chemistry to gain a mark.	2c Alternative Intermediate structure secondary carbocation OR 	

		Stage 3 Optical isomerism 3a 2-bromo isomer has chiral carbon / C with four different groups / non superimposable mirror images OR $\begin{array}{c} \text{CH}_3 \\ \\ \text{C} \cdots \cdots \text{CN} \\ \quad \diagup \\ \text{H} \quad \text{Br} \end{array}$ $\begin{array}{c} \text{CH}_3 \\ \\ \text{NC} \cdots \cdots \text{C} \\ \diagup \quad \\ \text{Br} \quad \text{H} \end{array}$ 3b Optical because (secondary) C⁺ planar 3c So can be attacked from above or below	
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Question	Answers	Additional Comments/Guidelines	Mark
06.3	M1 KCN or NaCN M2 Aqueous AND ethanol (alcohol)	Penalise acid in M1	2 (2 x AO1)

Question	Answers	Additional Comments/Guidelines	Mark
07.1	Use H_2SO_4	Allow HCl / H_3PO_4 Ignore conc / dilute	1 (AO1)

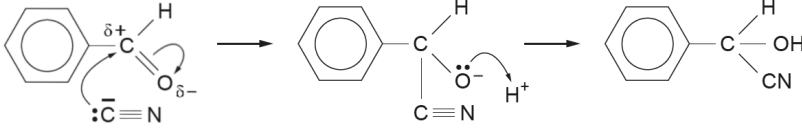
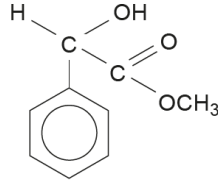
Question	Answers	Additional Comments/Guidelines	Mark
07.2	M1 Cool test 2 warm (water bath) M2 Gas is tested with lighted splint in test 3 Bubble into limewater	Allow heat / hot Allow no test on gas needed	2 (2 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
07.3	M1 J and M M2 Test 1 (Orange solution goes) green M3 M M4 Test 2 (Blue solution gives a brick) red precipitate M5 J and L M6 Test 3 (Colourless gas that turns) limewater cloudy M7 K M8 Test 4 (Orange solution goes) colourless	Allow (Brown-red/orange/orange-red) Allow M6 Test 3 fizz / effervescence Allow (Brown/Brown-red/yellow/yellow-orange) Allow decolorises bromine	8 (8 x AO3)

Question	Answers	Additional Comments/Guidelines	Mark
08.1	$C_nH_{2n-2}O$	Allow $C_nH_{2n}CO$ or $(CH_2)_nCO$ or $C_nH_{2(n-1)}O$	1 (AO2)

Question	Answers	Additional Comments/Guidelines	Mark
08.2	M1 curly arrow from bond to O M2 lone pair and curly arrow from lp to H M3 curly arrow from C-H bond to C-C bond	Allow other C-O bond breaking for M1	3 (3 x AO2)

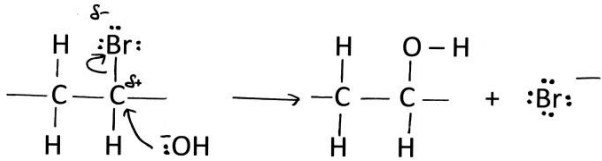
Question	Answers	Additional Comments/Guidelines	Mark
08.3	add $NaHCO_3$ / Na_2CO_3 (1) effervescence with the most acidic (1)		2

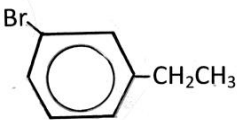
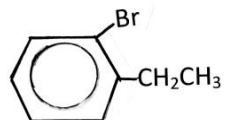
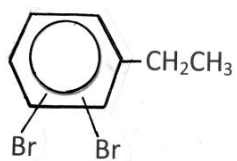
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)	I	 award (1) each for up to three of following curly arrows and addition of CN^- capture of H^+ partial and full charges dissociation of HCN into H^+ and CN^-		3		3		
			II	nucleophilic addition	1			1		
		(ii)		it is a racemic mixture / an equimolar mixture of the two enantiomers			1	1		
		(iii)		it does not contain a chiral centre / no asymmetric carbon atom		1		1		
		(iv)	I	it acts as a carboxylic acid 		1		2		
			II	it acts as an alcohol (giving an ester)		1		1		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		orange to green (accept colourless to green)	1			1		1
		(ii)		Indicative content - Dissolve the sodium benzoate in minimum amount of water - Filter the sand using a filter paper and transfer into a beaker - Add HCl into beaker, benzoic acid and sodium chloride will be formed - If benzoic acid is cooled, solubility of benzoic acid decreases - Description of recrystallization steps - Using data given in question to perform calculation		3	3	6		6
				5-6 marks All the essential steps described in logical order to ensure a dry sample of pentan-2-one <i>The candidate constructs a relevant, coherent and logically structured account including all key elements of the indicative content. A sustained and substantiated line of reasoning is evident and scientific conventions and vocabulary are used accurately throughout.</i> 3-4 marks Most of the correct steps described (although order may not be correct); safe method involving no naked flames <i>The candidate constructs a coherent account including many of the key elements of the indicative content. Some reasoning is evident in the linking of key points and use of scientific conventions and vocabulary is generally sound.</i> 1-2 marks Some correct steps in separation method <i>The candidate attempts to link at least two relevant points from the indicative material. Coherence is limited by omission and/or inclusion of irrelevant materials. There is some evidence of appropriate use of scientific conventions and vocabulary.</i> 0 marks <i>The candidate does not make any attempt or give an answer worthy of credit.</i>						

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
		(iii)		n = 0.0336 mol (1) Mr = 149 g mol ⁻¹		2		2		
		(iv)		alkaline iodine (iodoform) / KI and NaOCl (1) yellowprecipitate / solid forms (1) do not accept 2,4-DNPH test		2		2		2
				Question 9 total	3	12	5	20	0	9

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		(deep) blue Fehling's solution (1) gives a red / brown precipitate (1)	2			2		2
		(ii)		award (1) for name/formula of any aliphatic aldehyde e.g. propanal / CH ₃ CH ₂ CHO do not accept benzaldehyde / C ₆ H ₅ CHO		1		1		
	(b)	(i)		nucleophilic addition	1			1		
		(ii)		decomposition by water / dilute acid	1			1		
	(c)			award (1) each for any two of following loss of (volatile) ethanal incomplete oxidation / further oxidation side reactions			2	2		2
	(d)			moles of NaOH = $0.200 \times \frac{40.0}{1000} = 0.0080 \text{ mol}$ (1) moles of lactic acid reacted = 0.0080 mol mass of lactic acid reacted = $0.0080 \times 90.06 = 0.72 \text{ g}$ (1) volume of lactic acid = $\frac{0.72}{1.2} = 0.60 \text{ cm}^3$ percentage = $\frac{0.60}{75} \times 100 = 0.8\%$ (1)		2	1	3	1	

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
11	(a)		I	$H^+ / H^{\delta+}$		1		1		
			II	 curly arrows (1) partial charges (1)		1	1	2		
			III	if not present in excess the added sodium hydroxide will simply neutralise the two COOH groups present			1	1		1
			IV	the bromine atom has higher electronegativity and attracts both electrons from the C—Br bond (resulting in heterolytic fission)		1		1		
				Question 11 total	7	7	3	17	0	4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
12	(a)	(i)		bromine and iron(III) bromide / Br ₂ and FeBr ₃		1		1		1
		(ii)		17	1			1		1
		(iii)	I	award (1) for both of following structures   substitution can occur elsewhere in the ring (1)	1	1		2		
			II			1		1		
	(b)			alcoholic sodium hydroxide solution (1) a small molecule / HBr is removed (1)	1	1		2		2
	(c)			add aqueous bromine dropwise (1) bromine decolourised (1)	1	1		2		2

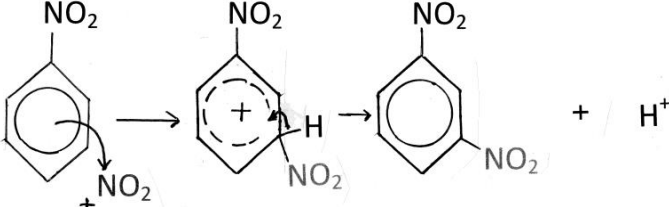
Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(d)			moles of phenylethene = $\frac{3.00}{104}$ = 0.289 mol moles of phenyloxirane = $\frac{2.50}{120}$ (1) = 0.208 mol mole ratio is 1:1 (1) percentage yield = $\frac{0.208}{0.289} \times 100 = 71.9\%$ (1)		3		3	2	
	(e)			a peak would be seen between 1650 and 1750 cm ⁻¹ (1) indicating the presence of the C=O group (1)	2			2		

Question				Marking details	Marks available																	
					AO1	AO2	AO3	Total	Maths	Prac												
	(f)			<table><tr><td>Test</td><td>phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$</td><td>phenylethanone $\text{C}_6\text{H}_5\text{C}(\text{O})\text{CH}_3$</td></tr><tr><td>alkaline iodine</td><td>no observable reaction</td><td>yellow precipitate formed</td></tr><tr><td>Tollens' reagent</td><td>silver mirror formed</td><td>no observable reaction</td></tr><tr><td>warming with acidified potassium dichromate</td><td>solution turns from orange to green</td><td>no observable reaction</td></tr></table> award (3) for all correct award (2) for any four or five correct award (1) for any two or three correct	Test	phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$	phenylethanone $\text{C}_6\text{H}_5\text{C}(\text{O})\text{CH}_3$	alkaline iodine	no observable reaction	yellow precipitate formed	Tollens' reagent	silver mirror formed	no observable reaction	warming with acidified potassium dichromate	solution turns from orange to green	no observable reaction		3		3		3
Test	phenylethanal $\text{C}_6\text{H}_5\text{CH}_2\text{CHO}$	phenylethanone $\text{C}_6\text{H}_5\text{C}(\text{O})\text{CH}_3$																				
alkaline iodine	no observable reaction	yellow precipitate formed																				
Tollens' reagent	silver mirror formed	no observable reaction																				
warming with acidified potassium dichromate	solution turns from orange to green	no observable reaction																				
	(g)			H H CH₃ HO—C—C—C—OH H H H			1	1														
				Question 12 total	6	12	2	20	2	10												

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
13		(i)	I	$\text{ClCH}_2(\text{CH}_2)_2\text{CH}_2\text{Cl} + 2\text{KCN} \rightarrow \text{NC}(\text{CH}_2)_4\text{CN} + 2\text{KCl}$		1		1		
			II	CN^-	1			1		
			III	award (1) for any of following lithium tetrahydridoaluminate(III) / LiAlH_4 hydrogen with Ni catalyst sodium and ethanol	1			1		1

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
14	(a)		I	award (1) for appropriate charges award (1) for curly arrows		2		2		
			II	initial attack is by the nucleophile CN^- (1) HCN adds across the C=O bond (1)	2			2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)	(i)		lithium tetrahydridoaluminate(III) / LiAlH_4 (1) hydrogen is added / oxygen is lost (1)	2			2		
		(ii)		benzene-1,4-dioic acid accept terephthalic acid		1		1		
	(c)	(i)		$ \begin{array}{c} \text{COO}^-\text{Na}^+ \\ \\ \text{CH}_2 \\ \\ \text{COO}^-\text{Na}^+ \end{array} + 2\text{NaOH} \longrightarrow 2\text{Na}_2\text{CO}_3 + \text{CH}_4 $		2		2		
		(ii)		sodium hydrogencarbonate / NaHCO_3 (1) accept sodium carbonate / Na_2CO_3 reaction complete when no more effervescence seen (1) phenol does not react with sodium hydrogencarbonate (to give carbon dioxide) (1)		3		3		3
		(iii)		no purple coloration seen			1	1		1
				Question 14 total	5	12	1	18	1	4

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
15		(i)				2		2		
		(ii)	I	Fe / Sn and hydrochloric acid	1			1		1
			II	NaOH / sodium hydroxide	1			1		1

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